



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2025-26

class: SYMCA	Div.: B	Course Code: MCA01604	Batch: S2
Semester: III		Course Name: Data Science Laboratory	
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CO No: CO605.1		Assignment No: 4	

Title: A company wants to predict the salary of employees based on their years of experience. The company has collected data on the salaries and experience of 100 employees. Use simple linear regression to model the relationship between salary and experience.

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

np.random.seed(0)
years_experience = np.random.uniform(1, 15, 100)
salary = 45000 + years_experience * 5000 + np.random.normal(0, 10000, 100)

X = years_experience.reshape(-1, 1)
y = salary

model = LinearRegression()
model.fit(X, y)

y_pred = model.predict(X)

plt.style.use('seaborn-v0_8-whitegrid')
plt.figure(figsize=(10, 6))

plt.scatter(X, y, color='blue', alpha=0.7, label='Actual Employee Salaries')
plt.plot(X, y_pred, color='red', linewidth=2, label='Predicted Salary Trend')

plt.title('Salary vs. Years of Experience')
plt.xlabel('Years of Experience')
plt.ylabel('Salary ($)')
plt.legend()
plt.grid(True)

plt.show()
```



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Output:-

